Regional heterogeneity in the second-hour tax

The Second-Hour Tax, PISA 2025 | Empirical supplement | 10 October 2026

Finding. The magnitude of the LDW second-hour penalty varies by region. The strongest main-sample contrast across alternative synthesis methods is East Asia versus Southern Europe. East Asia versus Western Europe is directionally consistent but fails correction across all regional comparisons.

The original archived estimates reproduce the paper: **84 systems, 205,114 students, 78 negative system effects**, and an international fixed-effect mean of -0.0931 within-system SD. The new analysis changes geographic synthesis, while retaining each system's original plausible-value and Fay-BRR uncertainty.

Regional LDW-second effects

Region	k	Students	Effect (SD)	95% CI
East Asia	7	15,246	-0.031	[-0.075, 0.014]
South-eastern Asia	7	18,134	-0.035	[-0.076, 0.006]
Central Asia	3	11,718	-0.046	[-0.106, 0.014]
Western Asia	11	31,702	-0.153	[-0.187, -0.119]
Western Europe	7	13,349	-0.101	[-0.145, -0.057]
Northern Europe	10	17,629	-0.085	[-0.127, -0.042]
Southern Europe	12	29,911	-0.123	[-0.157, -0.089]
Eastern Europe	8	15,235	-0.094	[-0.133, -0.055]
Northern America *	2	7,900	-0.113	[-0.198, -0.029]
Latin America and Caribbean	13	34,975	-0.092	[-0.123, -0.062]
Australia and New Zealand *	2	5,192	-0.087	[-0.169, -0.004]
Africa *	2	4,123	-0.078	[-0.164, 0.007]

k = education systems. Primary synthesis: one region intercept per group, REML residual between-system variance, modified Knapp-Hartung intervals with 72 degrees of freedom. More negative means a larger penalty. * Two-system groups borrow the common residual variance and have limited regional coverage. Africa combines Kenya and Morocco.

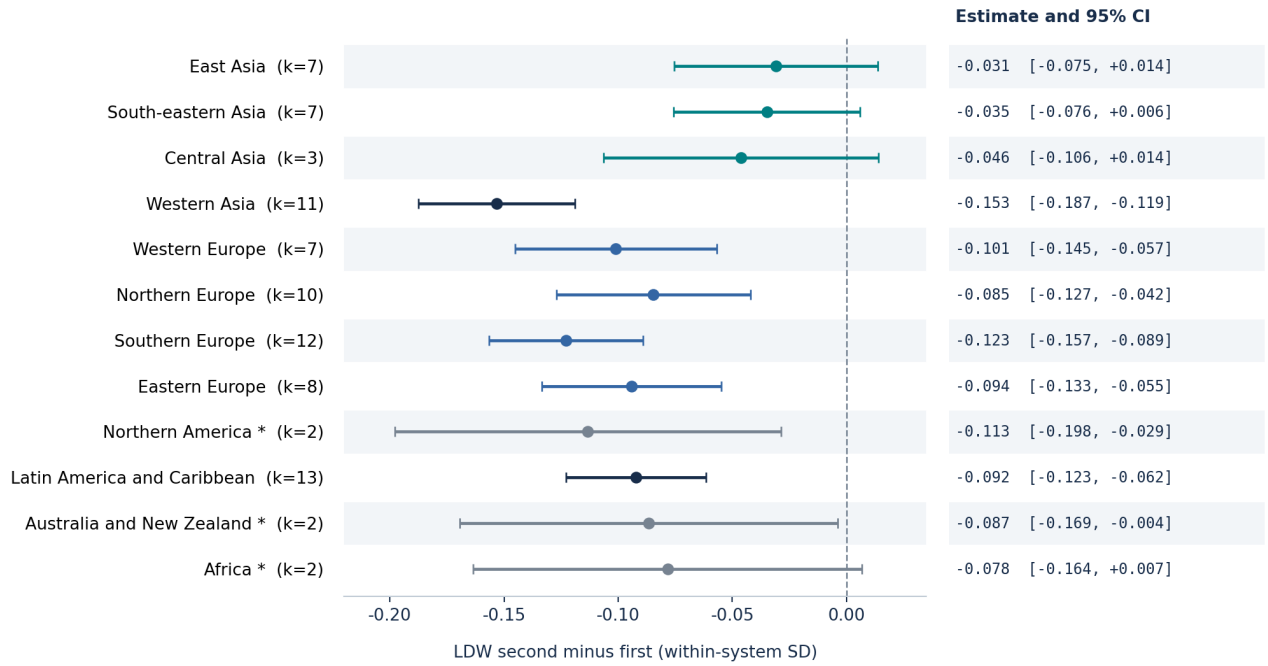
Interpretation. East Asian, South-eastern Asian and Central Asian averages are relatively close to zero, but their confidence intervals do not establish zero effects. Geography is an effect modifier, not a randomly assigned treatment.

The regional pattern

Primary model and explicit between-region tests

The second-hour penalty differs across regions

84 education systems | 205,114 students | Geography based on UN M49



Negative values indicate a larger LDW second-hour penalty.

REML regional meta-regression; modified Knapp-Hartung intervals. * Only two systems; sparse coverage.

Two comparisons of substantive interest

Contrast	Method	Difference (SD)	p	BH q
East Asia - Western Europe	REML / mKH	+0.070	0.0293	0.1623
East Asia - Western Europe	Separate-region DL / mKH	+0.071	0.0126	0.1219
East Asia - Western Europe	Equal-system / Welch	+0.070	0.0021	0.0622
East Asia - Southern Europe	REML / mKH	+0.092	0.0016	0.0264
East Asia - Southern Europe	Separate-region DL / mKH	+0.095	0.0007	0.0480
East Asia - Southern Europe	Equal-system / Welch	+0.091	0.0002	0.0152

Each BH correction uses the same complete family of 66 regional pairs. Positive contrasts mean a smaller penalty in East Asia. The primary East Asia - Southern Europe contrast is 0.092 SD, 95% CI [0.036, 0.148]. Its primary Holm-adjusted p is 0.101; the main claim is therefore about FDR-controlled exploratory evidence, rather than uniform familywise significance.

Five primary comparisons pass BH, but only East Asia versus Southern Europe also passes BH under both separate regional variance estimation and equal-system Welch inference. The strong East Asia versus Western Asia comparison in the common-variance model weakens when Western Asia's larger dispersion is allowed.

What survives the checks

Separate overall heterogeneity from certainty about a particular pair

Overall equality of regional means

Specification	Systems / regions	p
REML: all regions	84 / 12	0.0018
REML: at least two students per order	84 / 12	0.0061
REML: science-LDW pair only	84 / 12	0.0126
REML: omit Armenia and Lebanon	82 / 12	0.0031
REML: regions with at least three systems	78 / 9	0.0004
Welch: All 12 groups	84 / 12	0.0510
Welch: At least three systems per region	78 / 9	0.0016

The equal-system, unequal-variance test over all 12 groups is borderline (p=0.0510). Its degrees of freedom are constrained by the three two-system groups. With the pre-specified minimum of three systems per region, the nine-group Welch test gives p=0.0016. Regional heterogeneity is therefore supported in the adequately covered groups, while the sparse groups deserve descriptive interpretation.

East Asia versus Southern Europe: narrower samples

Specification	Difference	p	BH q
Primary main sample	+0.092	0.0016	0.0264
Science-LDW pair only	+0.086	0.0079	0.1082
At least two students per order	+0.086	0.0071	0.0782
Exclude Mongolia	+0.088	0.0036	0.0481

The contrast retains its direction under stricter support and a common paired domain, but fails their 66-comparison FDR correction. Across all 84 leave-one-system-out models its point estimate ranges from +0.084 to +0.102 SD, while BH q ranges from 0.005 to 0.089. It is promising evidence, not a universally significant result.

Influence and geographic choices

Omitting Armenia and Lebanon changes Western Asia's mean from -0.153 to -0.115 SD, yet the regional REML omnibus remains significant (p=0.0031). East Asia - Western Europe stays positive in every leave-one-system-out run (+0.060 to +0.080 SD); its unadjusted p can rise to 0.073. Singapore remains in South-eastern Asia in the primary classification, and an East Asia-plus-Singapore sensitivity is recorded separately.

Secondary evidence

Domain pairing and response/process outcomes

Does geography matter within the same mirrored pair?

Mirrored pair	LDW: p / BH q	Paired domain: p	Direct point gap: p / BH q
Science - LDW	0.0126 / 0.0226	0.4658	0.0057 / 0.0171
Reading - LDW	0.0021 / 0.0171	0.5262	0.0083 / 0.0186
Math - LDW	0.0203 / 0.0304	0.6727	0.0038 / 0.0171

BH uses the nine outcome-by-pair omnibus tests. There is regional variation in LDW penalties and direct LDW-minus-conventional point-gap effects in all three pairings. The tests do not detect regional heterogeneity in the conventional-domain gain when that domain moves earlier. A nonsignificant test does not establish equal effects. Reading and mathematics use 83 systems because the conventional-domain plausible values are unavailable in Uzbekistan.

Response and process effects by selected region

Region	Credit (pp)	Disengaged (pp)	Time (%)	Actions (%)
East Asia	-1.29	+1.52	-7.12	-1.53
South-eastern Asia	-0.43	+0.65	-9.95	-2.84
Western Europe	-2.81	+3.39	-13.99	-6.92
Southern Europe	-2.77	+4.68	-21.43	-10.34
Western Asia	-2.10	+5.37	-25.63	-13.03

Credit is normalized item credit; disengagement is the share with the released disengagement status. Time and actions are 100[exp(beta)-1] from log(1 + outcome), so the exact percentage interpretation concerns (1 + time) and (1 + actions). Full regional estimates and intervals are in the CSV outputs.

Five process-outcome omnibus tests

Outcome	p	BH q (five tests)
Normalized credit	<0.0001	<0.0001
Disengaged share	0.0001	0.0001
Not-reached share	<0.0001	<0.0001
Total time: log(1 + time)	<0.0001	<0.0001
Actions: log(1 + actions)	<0.0001	<0.0001

All five process outcomes show regional heterogeneity in the primary model. Nevertheless, none of the five East Asia - Western Europe process contrasts passes its five-test BH correction (q=0.070 to 0.972). These patterns characterize responses to test position; they do not identify a causal mediation pathway.

Geography and reproducibility

An explicit crosswalk for every participating system

Region	Participating systems
East Asia	Hong Kong (China), Japan, Korea, Macao (China), Mongolia, B-S-J-Z (China), Chinese Taipei
South-eastern Asia	Brunei Darussalam, Indonesia, Malaysia, Philippines, Singapore, Thailand, Viet Nam
Central Asia	Kazakhstan, Kyrgyzstan, Uzbekistan
Western Asia	United Arab Emirates, Armenia, Azerbaijan, Georgia, Israel, Jordan, Lebanon, Palestinian Authority, Qatar, Saudi Arabia, Turkiye
Western Europe	Austria, Belgium, Switzerland, Germany, France, Luxembourg, Netherlands
Northern Europe	Denmark, Estonia, Finland, United Kingdom, Ireland, Iceland, Lithuania, Latvia, Norway, Sweden
Southern Europe	Albania, Spain, Greece, Croatia, Italy, Kosovo, North Macedonia, Malta, Montenegro, Portugal, Serbia, Slovenia
Eastern Europe	Bulgaria, Czechia, Hungary, Moldova, Poland, Ukrainian regions (17 of 27), Romania, Slovak Republic
Northern America	Canada, United States
Latin America and Caribbean	Argentina, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, Guatemala, Mexico, Peru, Paraguay, El Salvador, Uruguay
Australia and New Zealand	Australia, New Zealand
Africa	Kenya, Morocco

M49 places Armenia, Azerbaijan, Georgia and Turkiye in Western Asia, and the Baltic systems in Northern Europe. Chinese Taipei is assigned to East Asia and Kosovo to Southern Europe for statistical geography. B-S-J-Z is Beijing, Shanghai, Jiangsu and Zhejiang rather than all mainland China. Ukrainian regions represent 17 of 27 regions. Oceania here contains only Australia and New Zealand.

Data level and model

This supplement uses archived system coefficients and total variances from replication version 1.1.0. It does not download student microdata or refit the student regressions. The primary model is **tau_hat(c) = mu(region(c)) + u(c) + e(c)**, with released Var[e(c)] and common REML Var[u(c)]. Modified Knapp-Hartung scales covariance by max{1, residual Q/(K-G)}. All regional contrasts retain their complete multiple-testing families. No two estimates from the same system are treated as independent in one model.

Run and sources

The accompanying package contains untouched estimate inputs and hashes, the geographic crosswalk, analysis code, all comparison and robustness tables, and this report. Run **python3 code/analyze_regions.py**, then **python3 code/make_deliverables.py**. Integrity checks reproduce original international summaries and sample totals.

United Nations Statistics Division, M49: unstats.un.org/unsd/methodology/m49/
OECD (2026), PISA 2025 Results, Volume I, Reader's Guide: PISA system names and codes
Original analysis: The Second-Hour Tax LaTeX manuscript and replication package version 1.1.0.